

Uranus

Uranus is the seventh planet from the Sun and the first to be discovered with a telescope, by William Herschel in 1781. It is an ice giant with an equatorial diameter of about 51,118 kilometres, four times that of Earth. Methane in the upper atmosphere absorbs red light and gives the planet its pale blue-green colour.

The most unusual thing about Uranus is its axial tilt of about 98 degrees. The planet effectively rotates on its side, so during its 84-year orbit each pole spends about 42 years in continuous sunlight followed by 42 years of darkness. The cause is unknown but a giant collision early in its history is the usual explanation.

Uranus has the coldest atmosphere of any planet, with a minimum recorded temperature of about minus 224 degrees Celsius, even though Neptune is farther from the Sun. It has 13 known faint, dark rings, discovered in 1977 when the planet passed in front of a star.

Its 27 known moons are named after characters from the works of William Shakespeare and Alexander Pope, including Titania, Oberon, Umbriel, Ariel and Miranda. Miranda has one of the most varied surfaces known, with cliffs up to 20 kilometres high. Voyager 2 remains the only spacecraft to have visited Uranus, flying past in January 1986.